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VoNet: Vehicle Orientation Classification using Convolutional Neural Network

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ABSTRACT

This paper presents a novel convolution neural network for classifying the orientation (or viewpoint) of a vehicle in a given image. Current equipping sensors in self-driving car is able to produce bounding box of vehicles in the proximity, but it does not recognize the viewpoint of them. Analyzing surrounding cars' direction in very complex environment has a significant role for autonomous driving. Utilizing nothing but a captured image, the purpose of this research is to classify viewpoint of vehicle: (1) front; (2) rear; (3) side; (4) front-side; and (5) rear-side. Deep convolutional neural network is used as the tool in performing classification task. The approach involves examining different CNN architectures using a large scale car dataset. In addition to that, the goal of the model is to be small and fast enough for limited hardware resource. We are able to achieve 95% accuracy, 57ms inference time on Nvidia GRID K520 GPU, and 1.6 MB Caffe model size.

CCS Concepts

• Computing methodologies → Object recognition.

Keywords

Autonomous vehicle; orientation classification; deep convolutional neural network

1. INTRODUCTION

Most existing computer vision techniques has always been using conventional hand-featured engineering in vehicle detection. In recent year, deep learning has shown breakthrough result in vision task which make almost all previous methods obsolete. Vehicle classification and detection are not new in vision community. However, the emergence of autonomous driving has encouraged a wider area of research in recent years. Despite massive amount of effort, it remains a challenge due to the nature of complexity. For instance, even just a slight change in environment, such as weather or terrain, would perplex the system. This is the reason why equipping crucial interoperating sensors is indispensable. Existing commercial auto-pilot system generally can be achieved on highways because of the road is well-maintained, predictable and constraint. Unfortunately, such system is not practical in

navigating in complex urban driving environment. To achieve autonomous vehicle that can operate in urban driving environment safely, combination of sensors, hardware and software platform are being researched and developed. In the 2007 DARPA Urban Challenge, Stanford group used lidar, radar, GPS, IMU, camera for mapping, localization, object recognition, planning trajectory, traffic light detection and generic sign detection [1]. Yet, the system was far from practicality. Although lidar system is robust for mapping environment, it is vulnerable to malicious attempt. For example, laser pointer can be used to confuse the sensor. Radar, responsible for long-range vehicle detection, does not produce many features and has trouble discerning metal objects which possibly produces error. Further research remains open for these challenges. On the other hand, camera, which is comparable to human's sense when driving, is one of the immediate solutions to above mentioned trade-offs. In addition to that, there are a lot of rich features that camera produces left unexplored as an example in Figure 1.



Figure 1. Equipping with radar, lidar and other sensors, the vehicle is able to detect surrounding vehicle effectively.

Although bounding boxes were produced, there is no information about orientation of vehicles which is essential when cruising in an even more complicated situation.

First of all, in this paper, a new convolution neural network for vehicle's viewpoint classification using a given image will be introduced. The motivation behind this research is the absence of the most important feature when detecting vehicle – its viewpoint. To human's sense, there are 2 ways to identify where the vehicle is heading: (1) track the motion of the object movement; and (2) spot its orientation or viewpoint. The former problem involves requirement of temporal information across frame and motion data. The latter problem is the one that the paper is trying to solve. It is believed that using motion data is enough to pin point the movement of vehicle, but, in more complicated environment, this may mislead the behavior of the vehicle. For instance, a car might move from a distant toward the sensor which can be inferred 2 scenarios either: (1) the car is accelerating toward you; or (2) the car is reversing toward you. Using radar or lidar to pinpoint this orientation classification problem is not trivial. It is recognized

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that current advance system using multiple sensors has already been proven to be fairly reliable, but this research wants to point out that some useful information can be further obtained from an image and can be used in redundant with other equipment to produce even more robust and reliable outcome.

Secondly, in general, developing advance deep convolutional neural network pay attention only on accuracy which ignore other aspect such as speed and power consumption. In addition to that, delivering the model over the network has to be efficient. Due to the requirement power-hungry computation, this paper will also be attentive to the number of parameters and operations, speed, as well as accuracy. The validity of this implementation will be tested mainly against autonomous driving. For instance, given the orientation of the vehicle, the car can effectively predict other vehicles' direction; thereby, prevent unwanted outcome.

The philosophy behind the design of the system are:

- Simplicity: due to the nature of the target problem, the network does not need to be complicated and too large because it has to be ready for integrating into larger system.
- Speed: the network has to be as simple as possible to be considerably fast without sacrificing accuracy. For it to be applicable for autonomous driving, speed is unquestionable to be one of the most significant factors.
- Accuracy: the system has to achieve high accuracy to be reliable enough for real implementation.
- Light-weight: this application will be highly likely involving embedded platform and mobile computing. Therefore, compressing a network which is small enough to fit in limited memory and to communicate easily over the Internet is crucial.

2. RELATED WORK

There has never been any attempt in producing identical or similar academic research. Nevertheless, it is still useful to point out similar experiment within similar area and important breakthrough that influence this research.

In other previous research, using camera to detect vehicle is mostly operated on highway or straight road does not feature high variance in vehicle image. In other words, camera sees only either the rear view or the front view of the vehicle ahead in the field of view. Due to car's low variant structure, traditional approach for detection were based on statistical methods as in [2]. One breakthrough generic detection method was proposed by cascaded classifier of Viola and Jones [3]. Such system is not applicable for self-driving because it is not scalable to possible variety of car including size, type, and direction, as well as complex environment. Since the successful evolution of deep learning, the characteristic of object detection has also evolved as in [4].

In 1990's, the very first successful resurrection of neural network was done by LeNet [5] when Yann LeCun built an architecture for digit classification. It took around another decade to see another development made by Alex Krizhevsky et al. AlexNet [6] was the winner of ImageNet ILSVRC challenge in 2012. The lag was mainly because of dataset availability and computational capability. Since that point, exponential improvement of convolutional neural network opens new possibilities. Szegedy et al. from Google introduced GoogLeNet [7] using Inception module. They concluded that CNN could reduce number of parameters and increase accuracy with deeper architecture and advance technique. Other architectures include VGGNet [8], ResNet [9], and new ones are being improved and developed every year. They have paved new area of computer vision and

produced state-of-the-art result that some approaches can even surpass human level.

There are always tradeoffs when it comes with great accuracy. Neural networks consume considerable amount of memory and intensive computation which give the implementation phase difficulty especially in embedded and mobile devices. Therefore, new developments recently aim to reduce the computation and energy consumption. SqueezeNet is able to achieve the same level of AlexNet's accuracy while trimming down parameters to 50x fewer and much smaller model size [10]. To go beyond that, many experiments were made to further compress the model without sacrificing accuracy. Low-rank approximation approach can be seen in the works by [11]-[14], mainly focus on reducing inference time at test phase. Other alternative approaches include hashing trick, vector quantization, circulant project, and pruning [15]-[18], which center around downsizing memory usage.

In vision task, existing successful systems attempt to detect and classify cars, that instead of figuring out hand-engineered feature descriptor, using deep learning by feeding the model thousands of training data. There is an application of CNN in self-driving which was "End to End learning for self-driving cars" [19]. The paper introduces a system that command steering using a single front-facing camera by inputting raw pixels into a convolutional neural network. Another notable application is in "An Empirical Evaluation of Deep Learning on Highway Driving" [20] which propose a system of deep learning in vehicle and lane detection. There are many more researches related to autonomous driving popping out every year.

To our knowledge, there has not been any attempt in identifying the viewpoint of cars using convolutional neural network. We will proceed by introducing dataset used, architecture design, and result of the experiment.

3. DATASET

"Comprehensive Cars" (CompCars) [21] is a large-scale and comprehensive car image dataset containing 208,826 images of 1,716 car models. The images were collected from two scenarios: web-nature and surveillance-nature. The dataset includes different car views, internal and external parts, and rich attributes. Besides CompCars, there is another publicly available dataset: "Cars" [22]. The Cars dataset contains 16,185 images of 196 classes of cars.

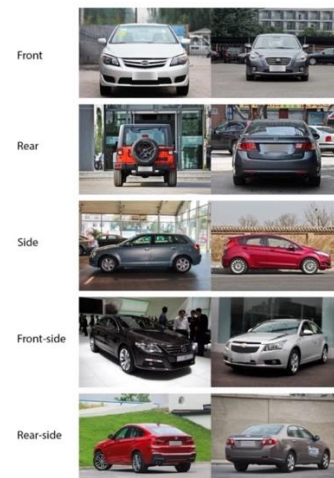


Figure 2. Samples of training dataset.

In this paper, "CompCars" will be chosen over "Cars" dataset due to its larger dataset for training and testing, containing diversely

distributed in all viewpoints and rich attribute annotations of each model. Although CompCars consists 208,826 images, our system will only use 16,016 images which capture the entire cars. The rest are car parts images and surveillance-nature data that are not applicable to this specific research. As the matter of fact, the most important notation in the dataset is car's viewpoint because they are target labels for training. To train the network, the training set is divided into 5 classes: (1) "Front" with 2,593 images; (2) "Rear" with 1,997 images; (3) "Side" with 2,973 images; (4) "Front-side" with 4,828 images; and (5) "Rear-side" with 3,625 images. All the images are high quality taken against different background and lighting as shown in Figure 2.

4. NETWORK ARCHITECTURE: VONET (VEHICLE ORIENTATION NET.)

Aforementioned, the method that will be used for classification task is deep convolutional neural network. Existing CNNs in image recognition include LeNet, AlexNet, GoogLeNet, VGG, ResNet and SqueezeNet. Comparison among these standard architectures will be conducted under the philosophy mentioned earlier for reference. LeNet will not be discussed because it does not fit in this application. VGG and Resnet produce high accuracy but their parameters and network is too large and overcomplicated that prolong computation time. Therefore, for this particular application, AlexNet, GoogLeNet and SqueezeNet will be used for experiment. On the top of that, this paper will propose a novel convolutional neural network called VoNet (Vehicle Orientation Net) that aims to classify vehicle orientation for high accuracy, relatively faster speed, and fewer parameters.

In this section, we will explain the philosophy behind VoNet architecture. The goal we are trying to achieve is to create a network that can achieve: (1) high accuracy; (2) fast inference; and (3) few parameters. Therefore, to achieve these goal, we take inspiration from GoogLeNet and SqueezeNet by picking the strong points from each model and construct a new architecture. GoogLeNet achieves incredibly high accuracy that use Inception module as the key; whereas SqueezeNet has the ability to shrink the network parameters and still achieve high performance by utilizing Fire module.

VoNet uses 2 main module: (1) I-module and (2) F-module. I-module consume high operations and require relatively more parameters if otherwise we insert it late in the model. This module is devised from Inception [4] by removing 1x1 and 5x5 convolutions. As experiment shows, by doing so, we can reduce number of computation and still preserve accuracy. The second module, F-module, is devised from Fire module [5]. Similar to the above alteration, the e1x1 of Fire module is remove due to the same reason. The module configuration is shown in Figure 3.

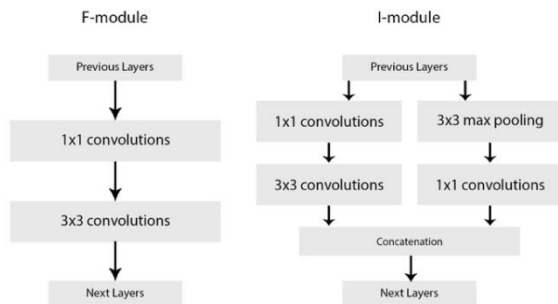


Figure 3. Two main modules used in VoNet architecture.

The rational is that the network can capture large area patch in early layer using 3x3 convolutions in I-module and let the later layers handle the smaller patch using 1x1 convolutions in F-module. 1x1 convolutions also help dimension reduction which results in less parameters. Moreover, we found that by not using fully-connected layer, the model can reduce the number of parameter and still can perform a superior result. Integrating both modules to work and complement each other, we can construct a CNN architecture as shown in Figure 4.

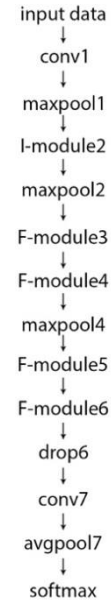


Figure 4. VoNet architecture.

5. NETWORK TRAINING

The weights are trained to minimize the error by comparing the 5 target labels and predicted labels using Caffe [23]. The algorithm used for back propagation is stochastic gradient descent (SGD). Weight initialization used is Xavier. Weight decay is 0.0005, momentum is 0.9 for faster learning rate, and dropout rate is 0.5 for preventing overfitting.

5.1 Train on AlexNet

From scratch. For reference, the network is trained from scratch using AlexNet architecture. ReLU nonlinearity, local response normalization, overlapping pooling, and dropout are some of the technique proposed to improve training phase and reduce overfitting. The base learning rate is 0.01. The input data is 227 x 227.

Partial fine-tune. Another experiment is to use the same set of training data on a pre-trained AlexNet. Partial fine-tune was done by removing the last fully connected layer and being replaced by a new one with 5 outputs. The network is then set the learning rate of all layer to 0 except the last one. The purpose is to maintain the feature that are already learned and iterate new data for only the new fully connected layer for learning. The main issue here is that dataset trained in AlexNet and CompCars does not match well, meaning that the dissimilarity of the dataset nature dilutes the effectiveness of the network.

5.2 Train on GoogLeNet

From scratch. Similar setting was done on GoogLeNet architecture. Comparing to AlexNet, GoogLeNet has larger depth

and less parameters of 4 million versus 60 million in AlexNet. The reason was that GoogLeNet, instead of using fully connected layer, contains only convolutional layers. However, the larger depth increases number of operations and feed forward time that works against our design goal. The input data is 224 x 224.

Partial fine-tune and full fine-tune. To go beyond learning the network from scratch, transfer learning of GoogLeNet architecture was done by replacing all the last classifier layers. In partial fine-tune, the learning rate of every layer are set to zero except the last ones; in full fine-tune, learning rate of every layer is left untouched.

5.3 Train on SqueezeNet

The configuration of SqueezeNet introduces new module called Fire module. The architecture claims to significantly reduce the number of parameters. Using CompCars dataset, training this network requires base learning rate as small as 0.001; otherwise, the model's accuracy does not improve and become totally useless. The input data is 227 x 227.

5.4 Train on VoNet

We constructed our new architecture and trained in similar condition from scratch with the base learning rate of 0.001 over 30 epochs. The input data is 227 x 227.

6. EXPERIMENT

Within the CompCars, the size of training set and validation set is 12,013 and 4,003 images, respectively. A single Nvidia GRID K520 GPU which has 4GB of memory was used for training.

Table 1. Performance comparison among all architecture (best result is represented in bold)

	Accuracy	Inference Speed (millisecond)	Size (MB)	Number of Parameters
AlexNet	0.9714	179.23	227	56,888,709
AlexNet Partial Fine-tune	0.9509	314.52		
GoogLeNet	0.9732	106.50	41.3	10,318,655
GoogLeNet Full Fine-tune	0.9770	106.47		
GoogLeNet Partial Fine-tune	0.9588	166.11		
SqueezeNet	0.9614	65.92	2.9	725,061
VoNet	0.9545	57.04	1.6	394,789

Table 1 shows summary of all experiment taken by comparing each model against their accuracy, speed, size, and parameters. The method in measuring the accuracy is by calculating accuracy in batch size of 25 and iterate through test set, then do the average on all results from every iteration. From the summary, the feed forward time of SqueezeNet run at an incredibly fastest speed comparing to the rest. It owes the fact that the model contains the least number of parameters, which also results in smaller model

size. On the top of that, the accuracy is fairly similar to other model that is still acceptable when put into use since there is only 0.03 accuracy differences from the best accuracy we can obtain. One interesting fact to be noted is that AlexNet and SqueezeNet can achieve accuracy almost the same as GoogLeNet due to the nature of the dataset. The 5 classes of the network, although similarity exists such as front versus rear image, are not complicated relative to ImageNet dataset.

Comparing to the rest, VoNet performs best in terms of inference speed, model size, and number of parameters. The major reason to this is the 2 newly introduced modules. The goal of creating the modules is to reduce number of parameter and operations; thereby, it reduces inference time and shrink the model size. However, it slightly hurts the accuracy. The sacrifice of tiny accuracy for speed and size is tolerable in this application.

7. DISCUSSION

What we can conclude from the error analysis are:

- Modern design of the car can obscure the network prediction.
- The dataset should pay attention more on angle of captured image which would affect accuracy considerably.
- Revision of the dataset annotation is needed.

One major drawback of this system is the inability to distinguish between right side and left side of the car due to the lack of notation in the dataset. Since the initial purpose of the dataset construction does not fully in parallel to this system, the network cannot be fed and differentiate between the two sides. Originally, the dataset is concentrate on fine-grained classification of car model. Therefore, this obvious mistake cannot fall on the dataset. To overcome this, revising the dataset notation whether each image is right or left side of the car and pivot the purpose to orientation classification will help the network to recognize different perspective (left, right, front-right, front-left, rear-right, and rear-left). Moreover, uniform distribution of training data will improve the accuracy in respective classes.

In autonomous vehicle, there are interoperating sensors which complement one another to give the car perception for making decision, as in [24]. Consequently, this proposed system is no different. One of the best method to improve it is to include the ability of other sensors to operate in conjunction to yield faster and better result. For example, with the help of lidar, the network can trim out some classes that is clearly not in the result (e.g. if the detected car's bounding box is square from the camera point of view, it must be either 'front' or 'rear', so the network can cut out the rest.)

Another aspect that we should be attentive to is real-time implementation. In video, the frame rate is usually 30 fps, so each frame duration is about 33.33ms. For this reason, the ideal calculation for the classification should to be completed in around 30ms. Nonetheless, in production implementation, there are other calculations needed, plus some extra overhead which make this ideal seem less realistic. Fortunately, considering the nature of application, identifying an orientation of a car does not require abrupt change. In most case, cars do not swiftly change its orientation within split millisecond. It would take 1-2 seconds for a complete switch.

8. CONCLUSION

The paper has solved the classification problem of identifying vehicle orientation utilizing CNNs by learning from data. The accuracy achieved is around 95% without manual feature

engineering or data augmentation. Testifying how well the overall result is following the philosophy of the system, it can be concluded that VoNet is the fittest network for this particular application which is simple, light-weight, accurate, and fast.

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